INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

End-Autumn Semester Examination 2023-24

Date of Examination: 21 November 2023 **Session:** AN **Duration:** 3 Hrs **Full Marks:** 100

Subject No. : AI61004 **Subject :** Statistical Foundations of AI/ML

Department/Center/School: Centre of Excellence in Artificial Intelligence

Specific charts, graph paper, log book etc., required

Special Instructions (if any) :

ANSWER ANY 5 QUESTIONS. ALL PARTS OF EACH QUESTION MUST BE ANSWERED TOGETHER

Q1.

A) You are given a 6-faced dice which you initially believe to be fully fair. You carry out 2 rounds of 10 throws each, where the outcomes (frequency of faces) are as below:

	#1	#2	#3	#4	#5	#6
Round 1	3	2	0	0	2	3
Round 2	2	1	1	0	4	2

Using suitable probability distributions, show how your belief about the fairness of the dice gets updated after each round. Show necessary derivations. **[10 marks]**

B) Given the dataset below, check which fits better – a Gaussian distribution, or a Gamma distribution. Use the Method of Moments to estimate the necessary parameters. **[10 marks]**

D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
23	14	7	3	9	39	10	1	1	4
D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
5	5	14	7	4	3	9	3	15	3

Q2.

A) Prove that the samples obtained by rejection sampling actually follow the target distribution. **[4 marks]**

B) Explain with a small example why samples drawn using Metropolis Algorithm (not Metropolis-Hastings) may not reflect the target distribution. **[4 marks]**

C) Consider the following function: f(x) = a*N(x,-5,25) + b*Gamma(x,10,3) + c*Beta(x,10,10) where x is defined over the whole real line. Consider that Gamma

PDF is 0 in (-infinity, 0) and Beta PDF is 0 outside of (0,1). For what values of (a,b,c) is this a valid PDF, and what is the mode of this PDF? **[2 marks]**

D) I want to draw samples from the above PDF (choose suitable values of a, b, c) using Metropolis-Hastings Algorithm. The proposal distribution is N(X(t), s), where X(t) is the last accepted sample. Illustrate two situations where i) having small value of 's' is useful, and ii) having large value of 's' is useful. **[6 marks]**

E) Provide an algorithm to draw samples from Gamma(n, 0.5) distribution ('n' is an integer) using Box-Muller transform [Hint: remember chi-square distribution]**[4 marks]**

Q3.

A) Consider a Hidden Markov Model with 3 possible latent state values, with Gaussian emission distribution with parameters (-3, 9), (0, 4), (3, 9). The state transition parameters follow the relation: $p(Z(t)=k \mid Z(t-1)=j) \propto 1/(|k-j|+1)$. All states are equally likely at t=1. You are provided with a short sequence of observations X=(0.8, -2, 1.5). Find the probability distribution of Z(3), using the observations and parameters. **[8 marks]**

B) Now consider an Order-2 Hidden Markov Mode where Z(t) depends on both Z(t-1) and Z(t-2), as $p(Z(t)=k \mid Z(t-1)=j, Z(t-2)=i) \propto 1/(|k-i|+|k-j|+1)$. All states are equally likely at t=1 and t=2. Consider an observation sequence X of length T. You want to estimate the possible values of the corresponding latent states using Gibbs Sampling. Provide the full algorithm, including the sampling steps for all variables, and how you will estimate the marginal distributions of each latent variable from the collected samples. **[8 marks]**

C) Now suppose that the mean parameters of the state-specific emission distributions are not known, but they are treated as IID latent variables (m1, m2, ..) following N(0, S). How will the Gibbs Sampling algorithm now change? **[4 marks]**

Q4.

A) Consider the following model: $Y_i \sim Ber(p)$, $Z_i \sim N(0,1)$, $X_i \sim N(W_k Z_i, s)$ where $k = Y_i$. We have 3 observations of X and Z: [X1=3.6 Y1=1, X2=-1.2 Y2=0, X3=4.2 Y3=1]
Consider the following 2 choices of parameters:
P1: [W0=-5, W1=5, s=9, p=0.5], P2: [W0=0, W1=4, s=25, p=0.8]
Which of these two sets of parameters fits the observations best? Note that we do not observe Z. **[8 marks]**

B) Explain the E-M algorithm to estimate the parameters of a generic latent variable model p(Z,X), where Z is the set of latent variables and X is the set of observed variables. Explain why it is guaranteed to converge to at least a local minima. **[4+8=12 marks]**

Q5

A) Provide an algorithm based on importance sampling to carry out the integration of $(\sin(x)/(\log(x)+1))*x^5*(1-x)^4$ over the interval (0,1). If your proposal distribution is N(0,1), which samples will have high "importance scores"?. **[6 marks]**

B) You have a dataset X of 10 binary observations. Your Null Hypothesis is that the Bernoulli parameter is 0.5, while the alternate hypothesis is that this parameter is below 0.5. Your test-statistic T(X) is the fraction of '1's, and your rejection region is defined by ($T(X)-0.5<\epsilon$). What will be possible values of ϵ, if you are to reject the null hypothesis at 5% level of significance? **[4 marks]**

C) In the above case, suppose you have obtained 6 heads. What will be the p-value if i) alternate hypothesis is that parameter is below 0.5, ii) alternate hypothesis is that parameter is different from 0.5? **[4 marks]**

D) You are provided with observations (x1, x2, ..., xN) which are all real numbers. You want to fit a Gaussian Mixture Model of K parameters on it. How will you evaluate the goodness of fit of this model on the data using chi-square test? Write the test statistic, including necessary expressions for how to calculate it. Also write the rejection criteria for the null hypothesis at level of significance α. **[6 marks]**

Q6.

A) Draw 10 samples X=[x1, ... x10] from N(0,5) using Box-Muller transformation. Draw 10 more samples Y=[y1, ... y10] from N(x1,5), N(x2,5) .... N(x10, 5). Use the following pseudorandom number sequence in (0, 99): **[4 marks]**
[14 79 36 10 85 71 93 26 21 47 64 4 18 95 52 59 31 3 84 19]

B) Construct a permutation test to check if X and Y follow same distribution. **[6 marks]**

C) Use Kernel Density Estimate to fit a PDF on Y using i) Parzen Window, ii) Gaussian Kernel. Calculate the fitted densities at Y=0 for both cases. **[6 marks]**

D) You have a sequence of observations [3.4, -1.5, 7.2, -2.8, -0.4, 6.5]. Would you like to model this with a single high-variance Gaussian, or a mixture of two low-variance Gaussians? Explain your answer using AIC and BIC. **[4 marks]**